SAGAR PUBLIC SCHOOL

Integrated Programme | Topic: Gravitation

PRACTICE SHEET 1

Acceleration due to Gravity, Gravitational Field and Variations

Time Allowed: 60 Mins **Total Questions: 19**

Name: _____ Class/Sec: _____ Date: _____

Instructions: Attempt all questions. Questions in the *one or more than one option is correct* category are marked accordingly. Options are labelled as in the source book (A–D or 1–4). The answer key and a self-assessment page follow the questions.

Q.1 At what altitude will the acceleration due to gravity be 25% of that at the earth's surface (given radius of earth is R) ?

(A) $R/4$ (B) R (C) $3R/8$ (D) $R/2$

Q.2 Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole ($d \ll R$). The value of d is

(A) $\dfrac{\omega^2 R^2}{g}$ (B) $\dfrac{\omega^2 R^2}{2g}$ (C) $\dfrac{2\omega^2 R^2}{g}$ (D) $\dfrac{\sqrt{Rg}}{g}$

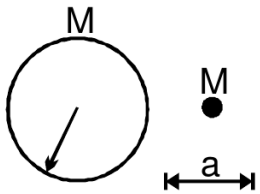
Q.3 If the radius of the earth be increased by a factor of 5, by what factor its density be changed to keep the value of g the same ?

(A) 1/25 (B) 1/5 (C) $1/\sqrt{5}$ (D) 5

Q.4 A man of mass m starts falling towards a planet of mass M and radius R. As he reaches near to the surface, he realizes that he will pass through a small hole in the planet. As he enters the hole, he sees that the planet is really made of two pieces: a spherical shell of negligible thickness of mass $2M/3$ and a point mass $M/3$ at the centre. Change in the force of gravity experienced by the man is

(A) $\dfrac{2}{3}\dfrac{GMm}{R^2}$ (B) 0 (C) $\dfrac{1}{3}\dfrac{GMm}{R^2}$ (D) $\dfrac{4}{3}\dfrac{GMm}{R^2}$

Q.5 A particle of mass M is at a distance a from surface of a thin spherical shell of equal mass and having radius a.



(A) Gravitational field and potential both are zero at centre of the shell.
(B) Gravitational field is zero not only inside the shell but at a point outside the shell also.
(C) Inside the shell, gravitational field alone is zero.
(D) Neither gravitational field nor gravitational potential is zero inside the shell.

Q.6 The escape velocity for a planet is v_e. A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

(A) v_e (B) $\frac{v_e}{\sqrt{2}}$ (C) $\frac{v_e}{2}$ (D) zero

Q.7 If a tunnel is cut at any orientation through earth, then a ball released from one end will reach the other end in time (neglect earth rotation)

(A) 84.6 minutes (B) 42.3 minutes

(C) 8 minutes (D) depends on orientation

Q.8 *(One or more than one option is correct)* Inside a hollow spherical shell

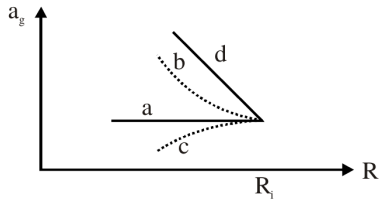
(A) everywhere gravitational potential is zero (B) everywhere gravitational field is zero

(C) everywhere gravitational potential is same (D) everywhere gravitational field is same

Q.9 The mass and diameter of a planet are twice those of earth. What will be the period of oscillation of a pendulum on this planet if it is a seconds pendulum on earth ?

(A) $\sqrt{2}$ second (B) $2\sqrt{2}$ seconds (C) $\frac{1}{\sqrt{2}}$ second (D) $\frac{1}{2\sqrt{2}}$ second

Q.10 A (non-rotating) star collapses onto itself from an initial radius R_i with its mass remaining unchanged. Which curve in figure best gives the gravitational acceleration a_g on the surface of the star as a function of the radius of the star during the collapse ?



(A) a (B) b (C) c (D) d

Q.11 Calculate the distance from the surface of the earth at which above and below the surface acceleration due to gravity is the same.

Q.12 The Earth may be regarded as a spherically shaped uniform core of density ρ_1 and radius $R/2$ surrounded by a uniform shell of thickness $R/2$ and density ρ_2. Find the ratio of $\frac{\rho_1}{\rho_2}$ if the value of acceleration due to gravity is the same at surface as at depth $R/2$ from the surface.

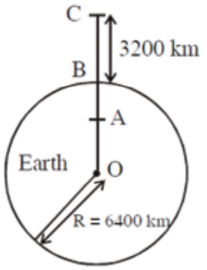
Q.13 The acceleration due to gravity on the earth's surface at the poles is g and angular velocity of the earth about the axis passing through the pole is ω. An object is weighed at the equator and at a height h above the poles by using a spring balance. If the weights are found to be same, then h is: ($h \ll R$, where R is the radius of the earth) **[JEE Main-2020]**

(1) $\frac{R^2\omega^2}{g}$ (2) $\frac{R^2\omega^2}{4g}$ (3) $\frac{R^2\omega^2}{2g}$ (4) $\frac{R^2\omega^2}{8g}$

Q.14 Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance $(R/2)$ from the earth's centre, where R is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period: **[JEE Main-2021]**

(1) $\frac{2\pi R}{g}$ (2) $\frac{g}{2\pi R}$ (3) $\frac{1}{2\pi}\sqrt{\frac{g}{R}}$ (4) $2\pi\sqrt{\frac{R}{g}}$

Q.15 In the reported figure of earth, the value of acceleration due to gravity is same at point A and C but it is smaller than that of its value at point B (surface of the earth). The value of OA : AB will be $x : y$. The value of x is ..... **[JEE Main-2021]**



Q.16 Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of the earth. If the escape speed on the surface of the earth is taken to be 11 $\mathrm{km\,s^{-1}}$, the escape speed on the surface of the planet in $\mathrm{km\,s^{-1}}$ will be: **[IIT-JEE 2010]**

Q.17 The temperature, at which the root mean square velocity of hydrogen molecules equals their escape velocity from the earth, is closest to: [Boltzmann Constant $k_B = 1.38 \times 10^{-23}$ J/K, Avogadro Number $N_A = 6.02 \times 10^{26}$ /kg, Radius of Earth: 6.4×10^6 m, Gravitational acceleration on Earth $= 10\ \mathrm{m\,s^{-2}}$] **[JEE Main-2019]**

(1) 800 K (2) 10^4 K (3) 650 K (4) 3×10^5 K

Q.18 A rocket has to be launched from earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, what should be the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon? Assume that the density of the earth and the moon are equal and that the earth's volume is 64 times the volume of the moon. **[JEE Main-2019]**

(1) $\frac{E}{4}$ (2) $\frac{E}{16}$ (3) $\frac{E}{64}$ (4) $\frac{E}{32}$

Q.19 A solid sphere of mass M and radius a is surrounded by a uniform concentric spherical shell of thickness $2a$ and mass $2M$. The gravitational field at distance $3a$ from the centre will be: **[JEE Main-2019]**

(1) $\frac{GM}{3a^2}$ (2) $\frac{2GM}{9a^2}$ (3) $\frac{GM}{9a^2}$ (4) $\frac{2GM}{3a^2}$

ANSWER KEY – PRACTICE SHEET 1

Answers are taken from the answer keys supplied with the source books (**Motion**: Answer_key.pdf; **Nucleus**: Answer Key at the end of GRAVITATION.pdf).

Q. No.	Source	Answer
1	Motion Ex-I Q.1	(B) R
2	Motion Ex-I Q.3	(A)
3	Motion Ex-I Q.4	(B)
4	Motion Ex-I Q.5	(A)
5	Motion Ex-I Q.7	(D)
6	Motion Ex-I Q.10	(B)
7	Motion Ex-I Q.12	(B)
8	Motion Ex-II Q.3	(B), (C), (D)
9	Nucleus O-1 Q.9	(B)
10	Nucleus O-1 Q.11	(B)
11	Nucleus S-1 Q.5	$\dfrac{\sqrt{5}-1}{2}R$
12	Nucleus S-1 Q.6	$\dfrac{7}{3}$
13	Nucleus JM Q.28	(3)
14	Nucleus JM Q.40	(4)
15	Nucleus JM Q.42	$x = 4$
16	Nucleus JA Q.3	$3\ \mathrm{km\,s^{-1}}$
17	Nucleus JM Q.14	(2)
18	Nucleus JM Q.15	(2)
19	Nucleus JM Q.16	(1)

Source abbreviations: Ex-I/II = Exercise I/II of the Motion booklet; O-1, S-1, JM, JA = Exercise (O-1), (S-1), (JM), (JA) of the Nucleus booklet.

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SELF-ASSESSMENT – PRACTICE SHEET 1

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Total Questions: 19	**Questions Attempted:** _____
Correct: _____	**Incorrect:** _____
Unattempted: _____	**Accuracy:** _____% (Correct / Attempted ×100)

Question-wise record (tick one: C = Correct, W = Wrong, S = Skipped)

Q.1 ☐C ☐W ☐S	**Q.2** ☐C ☐W ☐S	**Q.3** ☐C ☐W ☐S	**Q.4** ☐C ☐W ☐S
Q.5 ☐C ☐W ☐S	**Q.6** ☐C ☐W ☐S	**Q.7** ☐C ☐W ☐S	**Q.8** ☐C ☐W ☐S
Q.9 ☐C ☐W ☐S	**Q.10** ☐C ☐W ☐S	**Q.11** ☐C ☐W ☐S	**Q.12** ☐C ☐W ☐S
Q.13 ☐C ☐W ☐S	**Q.14** ☐C ☐W ☐S	**Q.15** ☐C ☐W ☐S	**Q.16** ☐C ☐W ☐S
Q.17 ☐C ☐W ☐S	**Q.18** ☐C ☐W ☐S	**Q.19** ☐C ☐W ☐S	

Reason for mistakes (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake

☐ Misread the question ☐ Ran out of time ☐ Guess work

Personal Feedback

What went well:

_____

_____

Topics / questions I need to revise:

_____

_____

My action plan before the next sheet:

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____